

LAB SHEET: 2

DML Commands – INSERT, UPDATE and DELETE

Perform the following SQL tasks using suitable examples:

1. Insert one record into the Student table.

```
1 • create database indus;
2   use indus;
3 • CREATE TABLE Student (
4       Student_ID INT PRIMARY KEY,
5       Student_Name VARCHAR(50),
6       Age INT,
7       City VARCHAR(30),
8       Course VARCHAR(50),
9       Fees INT
10  );
11 • INSERT INTO Student
12     (Student_ID, Student_Name, Age, City, Course, Fees)
13     VALUES
14     (101, 'Aman', 20, 'Meerut', 'B.Tech', 60000);
15 • select* from student;
16
```

Result Grid

	Student_ID	Student_Name	Age	City	Course	Fees
▶	101	Aman	20	Meerut	B.Tech	60000
*	NULL	NULL	NULL	NULL	NULL	NULL

2. Insert five student records using separate INSERT statements.

```
6 • INSERT INTO Student
7   VALUES (102, 'Ayesha', 20, 'Delhi', 'BCA', 45000);
8
9 • INSERT INTO Student
10  VALUES (103, 'Rohan', 21, 'Meerut', 'B.Tech', 60000);
11
12 • INSERT INTO Student
13  VALUES (104, 'Sakshi', 22, 'Noida', 'BBA', 40000);
14
15 • INSERT INTO Student
16  VALUES (105, 'Amit', 20, 'Agra', 'BCA', 45000);
17
18 • INSERT INTO Student
19  VALUES (106, 'Pallavi', 21, 'Ghaziabad', 'B.Tech', 60000);
20 • SELECT * FROM student;
21
```

Result Grid			Filter Rows:	Edit: 			Export
Student_ID	Student_Name	Age	City	Course	Fees		
102	Ayesha	20	Delhi	BCA	45000		
103	Rohan	21	Meerut	B.Tech	60000		
104	Sakshi	22	Noida	BBA	40000		
105	Amit	20	Agra	BCA	45000		
106	Pallavi	21	Ghaziabad	B.Tech	60000		

3. Insert multiple records into the Student table using a single statement.

```

INSERT INTO Student
(Student_ID, Student_Name, Age, City, Course, Fees)
VALUES
(107, 'Karan', 23, 'Meerut', 'BBA', 40000),
(108, 'Anjali', 21, 'Delhi', 'BCA', 45000),
(109, 'Vivek', 22, 'Agra', 'B.Tech', 60000),
(110, 'Riya', 20, 'Noida', 'BCA', 45000),
(111, 'Kunal', 21, 'Ghaziabad', 'BBA', 40000),
(112, 'Pooja', 22, 'Meerut', 'B.Tech', 60000),
(113, 'Deepak', 20, 'Agra', 'BCA', 45000),
(114, 'Nitin', 23, 'Delhi', 'BBA', 40000),
(115, 'Tanya', 20, 'Noida', 'B.Tech', 60000),
(116, 'Varun', 21, 'Ghaziabad', 'BCA', 45000),
(117, 'Komal', 22, 'Meerut', 'BBA', 40000),
(118, 'Aditya', 21, 'Delhi', 'B.Tech', 60000),
(119, 'Sneha', 20, 'Noida', 'BCA', 45000),
(120, 'Harsh', 23, 'Agra', 'BBA', 40000),
(121, 'Nisha', 21, 'Meerut', 'B.Tech', 60000),
(122, 'Mohit', 22, 'Delhi', 'BCA', 45000),
(123, 'Kavya', 20, 'Ghaziabad', 'BBA', 40000),
(124, 'Rahul', 21, 'Noida', 'B.Tech', 60000),
(125, 'Isha', 22, 'Meerut', 'BCA', 45000),
(126, 'Yash', 23, 'Delhi', 'BBA', 40000);
show tables;
select * from student;

```

107	Karan	23	Meerut	BBA	40000
108	Anjali	21	Delhi	BCA	45000
109	Vivek	22	Agra	B.Tech	60000
110	Riya	20	Noida	BCA	45000
111	Kunal	21	Ghaziabad	BBA	40000
112	Pooja	22	Meerut	B.Tech	60000
113	Deepak	20	Agra	BCA	45000
114	Nitin	23	Delhi	BBA	40000
115	Tanya	20	Noida	B.Tech	60000
116	Varun	21	Ghaziabad	BCA	45000
117	Komal	22	Meerut	BBA	40000
118	Aditya	21	Delhi	B.Tech	60000
119	Sneha	20	Noida	BCA	45000
120	Harsh	23	Agra	BBA	40000
121	Nisha	21	Meerut	B.Tech	60000
122	Mohit	22	Delhi	BCA	45000
123	Kavya	20	Ghaziabad	BBA	40000
124	Rahul	21	Noida	B.Tech	60000
125	Isha	22	Meerut	BCA	45000
126	Yash	23	Delhi	BBA	40000
131	Ayesha	20	Delhi	BCA	45000
NULL	NULL	NULL	NULL	NULL	NULL

4. Insert records into the Employee table.

```
> CREATE TABLE Employee (  
    Employee_ID INT PRIMARY KEY,  
    Employee_Name VARCHAR(50),  
    Salary DECIMAL(10,2),  
    Department VARCHAR(50)  
);  
  
INSERT INTO Employee  
(Employee_ID, Employee_Name, Salary, Department)  
VALUES  
(201, 'Rahul', 30000, 'IT'),  
(202, 'Vikas', 45000, 'HR'),  
(203, 'Pankaj', 25000, 'Sales'),  
(204, 'Sonia', 60000, 'IT'),  
(205, 'Meena', 35000, 'Finance'),  
(206, 'Amit', 28000, 'Sales'),  
(207, 'Neha', 50000, 'HR'),  
(208, 'Ravi', 40000, 'Finance');  
  
select * from student;
```

Employee_ID	Employee_Name	Salary	Department
201	Rahul	30000.00	IT
202	Vikas	45000.00	HR
203	Pankaj	25000.00	Sales
204	Sonia	60000.00	IT
205	Meena	35000.00	Finance
206	Amit	28000.00	Sales
207	Neha	50000.00	HR
208	Ravi	40000.00	Finance
NULL	NULL	NULL	NULL

5. Insert records into the Department table.

```
77 • CREATE TABLE Department (  
78     Department_ID INT PRIMARY KEY,  
79     Department_Name VARCHAR(50),  
80     Location VARCHAR(50)  
81 );  
82 • INSERT INTO Department  
83 (Department_ID, Department_Name, Location)  
84 VALUES  
85 (1, 'IT', 'Delhi'),  
86 (2, 'HR', 'Noida'),  
87 (3, 'Finance', 'Meerut'),  
88 (4, 'Sales', 'Ghaziabad'),  
89 (5, 'Marketing', 'Agra'),  
90 (6, 'Production', 'Delhi');  
91 • select * from Department;  
92
```

Result Grid			
Filter Rows:			
Edit:			
Department_ID	Department_Name	Location	
1	IT	Delhi	
2	HR	Noida	
3	Finance	Meerut	
4	Sales	Ghaziabad	
5	Marketing	Agra	
6	Production	Delhi	
NULL	NULL	NULL	

6. Insert records into the Product table.

```
94 • CREATE TABLE Product (  
95     Product_ID INT PRIMARY KEY,  
96     Product_Name VARCHAR(50),  
97     Price DECIMAL(10,2),  
98     Quantity INT  
99 );  
100 • INSERT INTO Product  
101     (Product_ID, Product_Name, Price, Quantity)  
102     VALUES  
103     (301, 'Laptop', 50000, 10),  
104     (302, 'Keyboard', 1500, 25),  
105     (303, 'Mouse', 800, 0),  
106     (304, 'Monitor', 12000, 5),  
107     (305, 'Printer', 15000, 0),  
108     (306, 'Headphone', 2000, 15),  
109     (307, 'Webcam', 3000, 8),  
110     (308, 'USB Cable', 500, 20);  
111 • select * from Product;  
112
```

Result Grid | Filter Rows: | Edit:

	Product_ID	Product_Name	Price	Quantity
	301	Laptop	50000.00	10
	302	Keyboard	1500.00	25
	303	Mouse	800.00	0
	304	Monitor	12000.00	5
	305	Printer	15000.00	0
	306	Headphone	2000.00	15
	307	Webcam	3000.00	8
	308	USB Cable	500.00	20
	NULL	NULL	NULL	NULL

7. Insert records into the Customer table.

```
112 • CREATE TABLE Customer (  
113     Customer_ID INT PRIMARY KEY,  
114     Customer_Name VARCHAR(50),  
115     City VARCHAR(30),  
116     Phone VARCHAR(15)  
117 );  
118 • INSERT INTO Customer  
119     (Customer_ID, Customer_Name, City, Phone)  
120     VALUES  
121     (401, 'Rakesh', 'Delhi', '9876543210'),  
122     (402, 'Suresh', 'Meerut', '9876543211'),  
123     (403, 'Nisha', 'Noida', '9876543212'),  
124     (404, 'Kavita', 'Delhi', '9876543213'),  
125     (405, 'Mohit', 'Ghaziabad', '9876543214'),  
126     (406, 'Pawan', 'Agra', '9876543215'),  
127     (407, 'Ritu', 'Meerut', '9876543216'),  
128     (408, 'Ankit', 'Noida', '9876543217');  
129 • select * from Customer;
```

Result Grid				
Filter Rows:		Edit:		
Customer_ID	Customer_Name	City	Phone	
401	Rakesh	Delhi	9876543210	
402	Suresh	Meerut	9876543211	
403	Nisha	Noida	9876543212	
404	Kavita	Delhi	9876543213	
405	Mohit	Ghaziabad	9876543214	
406	Pawan	Agra	9876543215	
407	Ritu	Meerut	9876543216	
408	Ankit	Noida	9876543217	
NULL	NULL	NULL	NULL	

8. Insert records into the Course table.

```
132 • CREATE TABLE Course (  
133     Course_ID INT PRIMARY KEY,  
134     Course_Name VARCHAR(50),  
135     Duration INT,  
136     Fees INT  
137 );  
138 • INSERT INTO Course  
139     (Course_ID, Course_Name, Duration, Fees)  
140     VALUES  
141     (501, 'B.Tech', 4, 60000),  
142     (502, 'BCA', 3, 45000),  
143     (503, 'BBA', 3, 40000),  
144     (504, 'MCA', 2, 70000),  
145     (505, 'MBA', 2, 80000),  
146     (506, 'M.Tech', 2, 75000),  
147     (507, 'B.Sc', 3, 35000),  
148     (508, 'M.Sc', 2, 55000);  
149 • select * from Course;
```

Result Grid				
		Filter Rows:		
	Course_ID	Course_Name	Duration	Fees
▶	501	B.Tech	4	60000
	502	BCA	3	45000
	503	BBA	3	40000
	504	MCA	2	70000
	505	MBA	2	80000
	506	M.Tech	2	75000
	507	B.Sc	3	35000
	508	M.Sc	2	55000
•	NULL	NULL	NULL	NULL

9. Insert a student record with values for all columns.

```
use indus;  
INSERT INTO Student  
(Student_ID, Student_Name, City, Course)  
VALUES  
(128, 'Ishita', 'Meerut', 'BCA');  
select * from student;
```

128	Ishita	NULL	Meerut	BCA	NULL
-----	--------	------	--------	-----	------

10. Update the city of a particular student.

```
158
159 • UPDATE Student
160 SET City = 'Delhi'
161 WHERE Student_ID = 107;
162 • SELECT * FROM Student
163 WHERE Student_ID = 107;
164
165
166
```

Result Grid | Filter Rows: Edit:

Student_ID	Student_Name	Age	City	Course	Fees
107	Karan	23	Delhi	BBA	40000

11. Update the salary of a particular employee.

```
.65 • UPDATE Employee
.66 SET Salary = 40000
.67 WHERE Employee_ID = 201;
.68 • SELECT * FROM Employee
.69 WHERE Employee_ID = 201;
```

Result Grid | Filter Rows: Edit:

Employee_ID	Employee_Name	Salary	Department
201	Rahul	40000.00	IT

12. Increase the salary of all employees by 10%.

```
71 • UPDATE Employee
72     SET Salary = Salary * 1.10;
73 • SELECT * FROM Employee;
74
75
```

Result Grid Filter Rows: Edit:			
Employee_ID	Employee_Name	Salary	Department
202	Vikas	45000.00	HR
203	Pankaj	25000.00	Sales
204	Sonia	60000.00	IT
205	Meena	35000.00	Finance
206	Amit	28000.00	Sales
207	Neha	50000.00	HR
208	Ravi	40000.00	Finance
NULL	NULL	NULL	NULL

13. Increase the price of all products by 5%.

```
76
77 • UPDATE Product
78     SET Price = Price * 1.05;
79 • SELECT * FROM Product;
80
81
82
```

Result Grid Filter Rows: Edit:			
Product_ID	Product_Name	Price	Quantity
303	Mouse	800.00	0
304	Monitor	12000.00	5
305	Printer	15000.00	0
306	Headphone	2000.00	15
307	Webcam	3000.00	8

14. Change the course of a particular student.

```
80
81 • UPDATE Student
82   SET Course = 'B.Tech'
83   WHERE Student_ID = 108;
84 • SELECT * FROM Student
```

result Grid						
Filter Rows:						
Student_ID	Student_Name	Age	City	Course	Fees	
108	Anjali	21	Delhi	B.Tech	45000	
NULL	NULL	NULL	NULL	NULL	NULL	

15. Update the department of an employee.

```
87
88 • UPDATE Employee
89   SET Department = 'IT'
90   WHERE Employee_ID = 203;
91 • SELECT * FROM Employee
92   WHERE Employee_ID = 203;
```

result Grid			
Filter Rows:			
Employee_ID	Employee_Name	Salary	Department
203	Pankaj	25000.00	IT
NULL	NULL	NULL	NULL

16. Update the city of all students from Meerut to Delhi.

```
196 • UPDATE Student
197     SET City = 'Delhi'
198     WHERE City = 'Meerut';
199
200 • SELECT * FROM Student;
201
202
203
204
205
```

Result Grid						
Filter Rows:						
Edit:   						
Student_ID	Student_Name	Age	City	Course	Fees	
101	Aman	20	Meerut	B.Tech	60000	
102	Ayesha	20	Delhi	BCA	45000	
103	Rohan	21	Meerut	B.Tech	60000	
104	Sakshi	22	Noida	BBA	40000	
105	Amit	20	Agra	BCA	45000	
106	Pallavi	21	Ghaziabad	B.Tech	60000	
107	Karan	23	Delhi	BBA	40000	
108	Anjali	21	Delhi	B.Tech	45000	
109	Vivek	22	Agra	B.Tech	60000	
110	Riya	20	Noida	BCA	45000	
111	Kunal	21	Ghaziabad	BBA	40000	
112	Pooja	22	Meerut	B.Tech	60000	

17. Update the fees of a particular course.

```
201
202 • UPDATE Course
203     SET Fees = 50000
204     WHERE Course_Name = 'BCA';
205 • SELECT * FROM Course
206     WHERE Course_Name = 'BCA';
```

Result Grid				
Filter Rows:				
Edit:   				
	Course_ID	Course_Name	Duration	Fees
▶	502	BCA	3	45000
•	NULL	NULL	NULL	NULL

18. Update multiple columns of a student simultaneously.

```
109 • UPDATE Student
110     SET Student_Name = 'Vivek Kumar',
111         City = 'Delhi',
112         Course = 'BCA',
113         Fees = 50000
114     WHERE Student_ID = 109;
115 • SELECT * FROM Student
```

Result Grid | Filter Rows: | Edit: | Export

Student_ID	Student_Name	Age	City	Course	Fees
109	Vivek Kumar	22	Delhi	BCA	50000
NULL	NULL	NULL	NULL	NULL	NULL

19. Delete a particular student record using Student_ID.

```
218 • DELETE FROM Student
219     WHERE Student_ID = 110;
220 • SELECT * FROM Student;
221
```

Result Grid | Filter Rows: | Edit: | Export

Student_ID	Student_Name	Age	City	Course	Fees
107	Karan	23	Delhi	BBA	40000
108	Anjali	21	Delhi	B.Tech	45000
109	Vivek Kumar	22	Delhi	BCA	50000
111	Kunal	21	Ghaziabad	BBA	40000
112	Pooja	22	Meerut	B.Tech	60000
113	Deepak	20	Agra	BCA	45000

20. Delete all students belonging to a particular city.

3

4 • USE indus;

5 • DELETE FROM Student

6 WHERE City = 'Meerut';

7 • SELECT * FROM Student;

Result Grid

Filter Rows:

Edit:

Exp

Student_ID	Student_Name	Age	City	Course	Fees
101	Aman	20	Meerut	B.Tech	60000
102	Ayesha	20	Delhi	BCA	45000
103	Rohan	21	Meerut	B.Tech	60000
104	Sakshi	22	Noida	BBA	40000
105	Amit	20	Agra	BCA	45000
106	Pallavi	21	Ghaziabad	B.Tech	60000
107	Karan	23	Delhi	BBA	40000

21. Delete employees whose salary is below a specified amount.

```
223 • DELETE FROM Employee
224   WHERE Salary < 30000;
225 • SELECT * FROM Employee;
226
```

Result Grid	Filter Rows:	Edit:	Exp
Employee_ID	Employee_Name	Salary	Department
201	Rahul	40000.00	IT
202	Vikas	45000.00	HR
203	Pankaj	25000.00	IT
204	Sonia	60000.00	IT
205	Meena	35000.00	Finance
206	Amit	28000.00	Sales
207	Neha	50000.00	HR
208	Ravi	40000.00	Finance
NULL	NULL	NULL	NULL

22. Delete products whose quantity is zero.

```

227 • DELETE FROM Product
228 WHERE Quantity = 0;
229 • SELECT * FROM Product;
230

```

Product_ID	Product_Name	Price	Quantity
301	Laptop	50000.00	10
302	Keyboard	1500.00	25
303	Mouse	800.00	0
304	Monitor	12000.00	5
305	Printer	15000.00	0
306	Headphone	2000.00	15
307	Webcam	3000.00	8
308	USB Cable	500.00	20
NULL	NULL	NULL	NULL

24. Delete records satisfying multiple conditions using AND/OR.

```

232 • DELETE FROM Student
233 WHERE (Age = 23 AND City = 'Delhi')
234 OR Course = 'BBA';
235 • SELECT * FROM Student;
236

```

Student_ID	Student_Name	Age	City	Course	Fees
101	Aman	20	Meerut	B.Tech	60000
102	Ayesha	20	Delhi	BCA	45000
103	Rohan	21	Meerut	B.Tech	60000
104	Sakshi	22	Noida	BBA	40000
105	Amit	20	Agra	BCA	45000
106	Pallavi	21	Ghaziabad	B.Tech	60000
107	Karan	23	Delhi	BBA	40000
108	Anjali	21	Delhi	B.Tech	45000
109	Vivek Kumar	22	Delhi	BCA	50000

25. Perform a complete sequence of **INSERT**, **UPDATE** and **DELETE** operations on a Student table and display the table after each operation.

248

249 -- UPDATE

250 • UPDATE Student

251 SET City = 'Noida',

252 Course = 'B.Tech',

253 Fees = 60000

254 WHERE Student_ID = 130;

255 • SELECT * FROM Student;

Result Grid | Filter Rows: | Edit: | Export/Import

Student_ID	Student_Name	Age	City	Course	Fees
101	Aman	20	Meerut	B.Tech	60000
102	Ayesha	20	Delhi	BCA	45000
103	Rohan	21	Meerut	B.Tech	60000
104	Sakshi	22	Noida	BBA	40000
105	Amit	20	Agra	BCA	45000
106	Pallavi	21	Ghaziabad	B.Tech	60000
107	Karan	23	Delhi	BBA	40000
108	Anjali	21	Delhi	B.Tech	45000

238

239 • USE indus;

240 -- INSERT

241 • INSERT INTO Student

242 (Student_ID, Student_Name, Age, City, Course, Fees)

243 VALUES

244 (130, 'Aman Kumar', 21, 'Delhi', 'BCA', 45000);

245 • SELECT * FROM Student;

Result Grid | Filter Rows: | Edit: | Export/Import

Student_ID	Student_Name	Age	City	Course	Fees
101	Aman	20	Meerut	B.Tech	60000
102	Ayesha	20	Delhi	BCA	45000
103	Rohan	21	Meerut	B.Tech	60000
104	Sakshi	22	Noida	BBA	40000
105	Amit	20	Agra	BCA	45000
106	Pallavi	21	Ghaziabad	B.Tech	60000
107	Karan	23	Delhi	BBA	40000
108	Anjali	21	Delhi	B.Tech	45000

```

258      -- DELETE
259 •    DELETE FROM Student
260      WHERE Student_ID = 130;
261 •    SELECT * FROM Student;

```

Result Grid						
		Filter Rows:		Edit:		Export/Import:
Student_ID	Student_Name	Age	City	Course	Fees	
101	Aman	20	Meerut	B.Tech	60000	
102	Ayesha	20	Delhi	BCA	45000	
103	Rohan	21	Meerut	B.Tech	60000	
104	Sakshi	22	Noida	BBA	40000	
105	Amit	20	Agra	BCA	45000	
106	Pallavi	21	Ghaziabad	B.Tech	60000	
107	Karan	23	Delhi	BBA	40000	
108	Anjali	21	Delhi	B.Tech	45000	